



Machine Learning and Pattern Recognition

Prof. Dr. Mohammed Elmogy

Mansoura University
Faculty of Computers and Information
Information Technology Department
melmogy@mans.edu.eg

27. September 2025



Outline

Course Objectives

Topics

Text books

Practical Part

Grading

Lecture Rules



Outline

Course Objectives

Topics

Text books

Practical Part

Grading

Lecture Rules



Course Objectives

- ▶ Introducing students to the **basic concepts** and **techniques** of **Machine Learning (ML)**.
- ▶ **Learning ML by doing it**, i.e. developing skills of using ML algorithms for solving practical problems.
- ▶ To gain experience of **doing independent study and research**.



Outline

Course Objectives

Topics

Text books

Practical Part

Grading

Lecture Rules



Topics

- ▶ What ML is, what **problems** it tries to solve, and the **main categories** and **fundamental concepts** of its systems.
- ▶ The **steps** in a typical ML project.
- ▶ Learning by **fitting a model to data**.
- ▶ **Optimizing** a cost function.
- ▶ **Handling, cleaning, and preparing** data.



Topics (cont'd)

- ▶ Selecting and engineering features.
- ▶ Selecting a model and tuning hyperparameters using cross-validation.
- ▶ The challenges of ML, in particular underfitting and overfitting.

Topics (cont'd)

- ▶ The **most common learning algorithms**: **Linear and Polynomial Regression, Logistic Regression, k-Nearest Neighbors, Support Vector Machines, Decision Trees, Random Forests, and Ensemble methods.**
- ▶ **Reducing the dimensionality** of the training data to fight the “curse of dimensionality”.
- ▶ Other **unsupervised learning techniques**, including **clustering, density estimation, and anomaly detection.**



Outline

Course Objectives

Topics

Text books

Practical Part

Grading

Lecture Rules

Textbooks

Hands-on ML

Auretien Geron. “Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems”, Second Edition, O'REILLY, 2019.

Additional Books

- ▶ Joel Grus's *Data Science from Scratch* (O'Reilly).
- ▶ Stephen Marsland's *Machine Learning: An Algorithmic Perspective* (Chapman & Hall).
- ▶ Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin's *Learning from Data* (AMLBook)

Textbooks

Hands-on ML

Auretien Geron. “Hands-on Machine Learning with Scikit-Learn, Keras & TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems”, Second Edition, O'REILLY, 2019.

Additional Books

- ▶ Joel Grus's **Data Science from Scratch** (O'Reilly).
- ▶ Stephen Marsland's **Machine Learning: An Algorithmic Perspective** (Chapman & Hall).
- ▶ Yaser S. Abu-Mostafa, Malik Magdon-Ismail, and Hsuan-Tien Lin's **Learning from Data** (AMLBook)



Additional Materials

YouTube AI Channel

Andrew Ng's ML course on Coursera.



Outline

Course Objectives

Topics

Text books

Practical Part

Grading

Lecture Rules



Practical Part

Python Book

Hetland, Magnus Lie: Beginning Python From Novice to Professional, Apress, 3rd ed., 2017.

- ▶ Python basics
- ▶ Implementing ML techniques



Outline

Course Objectives

Topics

Text books

Practical Part

Grading

Lecture Rules



Grading

- ▶ 20% Section Work + Practical Exam
- ▶ 20% Midterm
- ▶ 20% Assignments + Reports + Quizzes
- ▶ 40% Final Examination



Outline

Course Objectives

Topics

Text books

Practical Part

Grading

Lecture Rules

Lecture Rules

